USN | | | | | | | | | | [22CE42]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

FLUID MECHANICS AND HYDRAULICS

Time: 03 HOURS **Maximum Marks:50**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	**05 x02=10**		
1.	**a.**	What is a manometer? How are they classified?	L2	CO1	**02**
	b.	An open tank contains water up to a depth of 2m and above it an oil of specific gravity 0.9 for a depth of 1m. find the pressure intensity at the bottom of the tank.	2	2	**02**
	c.	Differentiate between uniform and non-uniform flow.	2	3	**02**
	d.	What do you understand by notches and weirs?	2	4	**02**
	e.	Explain the principle of dimensional homogeneity	2	6	**02**
		PART – B	**06 X 05=30**		
2.	**a.**	Define the terms 'gauge pressure', 'Vacuum pressure' and 'absolute pressure'. Indicate their relative position on chart.	2	1	**05**
		OR			
	b.	What is the advantage of single column manometer over the U-tube manometer? Briefly explain the working of single column manometer.	2	1	**05**
3.	**a.**	State and prove hydrostatic law.	2	2	**05**
		OR			
	b.	A circular plate 3m diameter is immersed in water in such a way that its greatest and least depth below the free surface are 4m and 1.5m respectively. Determine the total pressure on one face of the plate.	2	2	**05**
4.	**a.**	A 25 cm dia pipe carries oil of SG 0.9 at a velocity of 3m/s, at another section the diameter is 20cm. Find the velocity at this section and also mass rate of flow of oil.	2	3	**05**
		OR			
	b.	In a two-dimensional incompressible flow, the fluid velocity components are given by u=x-4y and v= -y-4x. Show that velocity potential exists and determine its form.	2	3	**05**
5.	**a.**	State Bernoulli's principle. List the assumptions which are made while deriving Bernoulli's equation.	3	4	**05**
		OR			

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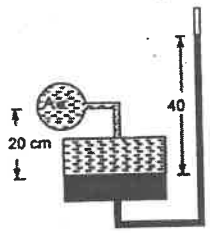
	b.	Crude oil of kinematic viscosity 0.4 stoke is flowing through a pipe of diameter 300mm at the rate of 300 L/s. Find the head lost due to friction for a length of 50m of the pipe	3	4	05
6.	a.	Discuss the different classification of the hydraulic turbines.	3	5	05
		OR			
	b.	Explain the general layout of a hydraulic power plant with a neat sketch.	3	5	05
7.	a.	Explain the different types of similarities that must exist between a prototype and its model.	3	6	05
		OR			
	b.	State and explain Buckingham π-theorem for dimensional analysis.	3	6	05
		PART – C	**06 X 10=60**		
8.	a.	Differentiate between Piezometer and U-tube manometer with sketches. A U-tube manometer tube is filled with two types of liquids. The density of liquid 1 (in left limb) is ρ_1=0.8 g/cm^3, and the density of liquid 2(in right limb) is ρ_2= 1g/cm^3, and height h_1 in left limb is 10 cm. What is the height of h_2 of liquid 2 in right limb?	3	1	10
		OR			
	b.	A single column manometer is connected to a pipe containing a liquid of specific gravity 0.9 as shown in figure 8(b). Find the pressure in the pipe if the area of the reservoir is 100 times the area of the tube for the manometer reading shown in fig. The specific gravity of mercury is 13.6  **Figure 8(b)**	3	1	10
9.	a.	Derive an expression for the total pressure on a submerged vertical plane surface by the static liquid and locate the position of centre of pressure.	3	2	10
		OR			
	b.	A vertical gate closes a horizontal tunnel 5m high and 3m wide running full of water. The pressure at the bottom of the gate is 196.2kN/m^2. Determine the total pressure on the gate and the centre of pressure.	3	2	10
10.	a.	Derive Continuity Equation for Three-Dimensional flow.	3	3	10

		OR			
	b.	Define and explain the terms 'Velocity potential function' and 'Stream function'	3	3	**04**
	c.	A stream function is given by $\psi=5x-6y$. Calculate the velocity components and also the magnitude and direction of the resultant velocity at a point.	3	3	**06**
11.	**a.**	What is meant by equivalent diameter of pipes in series and parallel? Derive expression for flow through pipes in series and pipes in parallel.	3	4	**10**
		OR			
	b.	Describe components and working of Venturimeter with a neat sketch. Derive the expression for discharge through it.	3	4	**10**
12.	**a.**	Explain Pelton turbine with a neat sketch	3	5	**10**
		OR			
	b.	Describe draft tube and its functions with neat sketch. What are the different types of draft tubes?	3	5	**10**
13.	**a.**	The efficiency η of a fan depends on the density ρ, the dynamic viscosity μ of the fluid, the angular velocity ω, diameter D of the rotor and the discharge Q. Express η in terms of dimensionless parameters.	3	6	**10**
		OR			
	b.	The resisting force F of a plane during flight can be considered as dependent upon the length of aircraft l, velocity v, air viscosity μ, air density ρ, and bulk modulus of air K. Express the functional relationship between these variables and the resisting force using dimensional analysis.	3	6	**10**